



United International University

Department of Computer Science and Engineering

CSE 3313: Computer Architecture

Midterm Examination: Summer 2025

Total Marks: 30 Time: 1 hour and 30 minutes

Any examinee found adopting unfair means will be expelled from the trimester / program as per UIU disciplinary rules.

Answer all questions. Numbers to the right of the questions denote their marks.

1. Consider computer A with a CPU speed of 2 GHz. The program P which has number of instruction and CPI as shown in Table 1.

Instruction	Count	CPI
addi	4	2
add	4	1
beq	2	4
bne	1	4
slt	2	3
sll	2	4
and	3	2
j	2	8

Table 1: Program P's number of instructions and CPI in Computer A architecture

Consider that Program P requires 50 ns to be completed Computer A.

- Find the execution time of program P in Computer A. [2]
 - Computer B with a CPU speed of 10 GHz has the execution time of the program P which is now 2 times faster than computer A and same instruction set architecture (ISA), calculate the average CPI of computer B. [2]
 - As we get the total execution time of the given program for Computer A from question (i), assume that the Arithmetic unit takes 65%, the logical unit takes 25%, and the branch operation takes 10% time of total execution time. Find the improvement factor of the given program if we replace the arithmetic unit with a better Arithmetic unit, which improves total completion time by 2x. [6]
2. (a) Consider the following C function. The arguments will be found at \$a0-\$a2. The compiler will assign a at \$s0. Convert the code to the corresponding MIPS assembly instructions. [8]

```
1  int my_function(int i, int n, int arr[])
2  {
3      int a = 10;
4      for(i = 0; i <= n; i++)
5      {
6          arr[ i * 2 ] = ~ (a & n);
7      }
8      return a;
9  }
```

Code for 2 (a)

- (b) Observe the following MIPS code and convert it into machine code. Conversion of the machine code to binary is not required. [3]

```
1  100: LOOP
2  104: slt $t3, $t5, $s1
3  108: bne $t3, $zero, REST
4  112: sw $t5, 20 ( $s4)
5  116: j LOOP
6  120: andi $t2, $s3, 30
7  124: REST
8  128: sub $t2, $a0, $s3
```

Code for 2 (b)

- (c) The students of UIU have designed a modified instruction set architecture called 'MIPS_3313' that follows the principles of the MIPS architecture but only with a few changes. The processor of MIPS_3313 has **16 registers**. The architecture is **byte-addressable** and the size of the instruction and register is **16 bits**. **3** bits are reserved for opcode. The structure for lw instruction in MIPS_3313 is given in the table-2. **Calculate** the maximum size of the array that the MIPS_3313 instruction can operate with. [4]

opcode	rs	rt	constant
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Table 2: Structure of lw instruction

3. (a) Multiply two numbers X by Y using the optimized multiplication algorithm, where $X = 5$ and $Y = 6$. [4]
 (b) You need to compute $P = A \times B \times C$ using unsigned binary integers. Let A be 9 bits wide, B be 14 bits wide, and C is defined as $C = A \times B$. What is the minimum width (in bits) required for the product register to hold P at any time? [1]

Instruction	Opcode	Function Code
add	0	32
sub	0	34
lw	35	-
sw	43	-
and	0	36
or	0	37
nor	0	39
andi	12	-
ori	13	-
sll	0	0
srl	0	2
beq	4	-
bne	5	-
slt	0	42
j	2	-
jr	0	8
jal	3	-
addi	8	-

(a) MIPS Machine Codes

Name	Register Number
\$zero	0
\$at	1
\$v0-\$v1	2-3
\$a0-\$a3	4-7
\$t0-\$t7	8-15
\$s0-\$s7	16-23
\$t8-\$t9	24-25
\$k0-\$k1	26-27
\$gp	28
\$sp	29
\$fp	30
\$ra	31

(b) MIPS Registers